

Volume Profile / Market Profile

Quantitative backtesting study — in search of a real edge, validated out-of-sample

33 years of data · 5 instruments (US + Brazil) · 4 strategies · walk-forward validation · costs included

Daily period	1993–2026 (as far back as each asset's history goes)
Instruments	SPY, QQQ (US) · PETR4, VALE3, BOVA11 (B3)
Data source	Yahoo Finance (yfinance), free
Strategies	POC reversion · Edge-to-Edge · VA breakout · Volume exhaustion
Key metric	Per-trade expectancy after costs
Validation	Walk-forward (optimize on the past, measure on the unseen future)
Generated on	June 2026

Technical research document. Not investment advice. Leveraged trading carries the risk of total loss.

1. Executive summary

We tested the objective rules of the Volume Profile strategy (POC, Value Area, day-types, the 80% Rule, edge-to-edge and volume reading) over 33 years of data, with costs and — crucially — with **walk-forward validation**: the parameters are chosen using only past data and measured on data the optimizer never saw. The central conclusions:

- **A real but modest edge exists on the long side of liquid indices.** Out-of-sample, QQQ and SPY are profitable across all four strategies (QQQ Edge-to-Edge: Profit Factor 2.06; QQQ Exhaustion: expectancy 1.80%/trade).
- **'Volume reading' works better than the profile itself.** Buying exhaustion ('no-supply' — new lows on below-average volume) was the most robust tactic out-of-sample on index ETFs (PF 1.5-2.2).
- **Reversion fades balanced markets well and fails in trend.** The Brazilian stocks (PETR4, VALE3), more trending/volatile, produced negative expectancy on almost everything — exactly what theory predicts.
- **Two of the method's 'legends' do not hold up in the data:** the 80% Rule did not traverse 80% of the time (27-67% in the recent intraday sample) and was negative after costs; and day-types do not predict continuation — if anything, 'bearish' days bounce more (reversion).
- **As a standalone system, the edge does not beat buy & hold on return** (low exposure), but it delivers a smoother stream: the diversified portfolio (OOS-validated sleeves) returned CAGR 1.4% with a maximum drawdown of only -6.7% and Sharpe 0.54.
- **Under rigorous falsification, the edge does not hold up as a standalone strategy.** Volume permutation, price-only ablation, random-entry control, excess return and bootstrap (§11) show that the gain is, to a large extent, **long exposure** to assets that went up. No sleeve generates alpha over its own exposure or beats the risk-free rate at 1% risk, and none has a Profit Factor whose 95% CI excludes 1.0. Only on SPY is the volume signal statistically real — though small and fragile at this sample size.

In one sentence: Volume Profile is a legitimate **context lens** and the volume filter **adds real selectivity**, but we found no **robust, standalone economic edge** — it is not the 'secret formula' sold in videos. The value lies in reading context and in risk management; operational use demands skepticism, cheap execution and modest expectations.

2. Methodology

2.1 Data and instruments

Ticker	Name	Market	Start	Sessions	Buy&Hold CAGR
SPY	SPDR S&P 500 ETF	US	1993-01-29	8,396	8.8%
QQQ	Invesco Nasdaq-100 ETF	US	1999-03-10	6,854	10.2%
PETR4	Petrobras PN	BR	2000-01-03	6,487	7.7%
VALE3	Vale ON	BR	2000-01-03	6,510	12.5%
BOVA11	iShares Ibovespa ETF	BR	2009-04-17	4,222	7.8%

Daily OHLCV data from Yahoo Finance (long history, reliable volume on ETFs/stocks). Free intraday data is short (~60 days of 30-min bars), which limits a faithful test of the intraday rules (80% Rule) to a recent sample — hence the **hybrid** approach: a long daily backtest for statistical robustness + recent intraday validation.

2.2 The four strategies

Strategy	Logic
POC reversion	Fades extremes: sells above the VAH / buys below the VAL, targeting the POC. The classic rotation of a balance.
Edge-to-Edge	Enters at one edge of the Value Area and aims for the opposite edge, crossing the interior of the profile. Large
VA breakout	Trades in the direction of a Value Area breakout (initiating activity), with an ATR-multiple target. Captures tren
Volume exhaustion	Buys 'no-supply' (\$6): new lows on below-average volume = exhausted sellers. Does not depend on the profil

2.3 Costs, sizing and assumptions

- **No lookahead:** profile levels use only data prior to the evaluated day; a signal at the close of day $t \rightarrow$ entry at the open of $t+1$.
- **Costs:** US ~0.05% round-trip (ETF + slippage); B3 ~0.2% (fees + slippage). All results are net.
- **Risk-based sizing:** each trade risks 1% of capital at the stop (ATR). Exits by stop, target or time; when ambiguous, the stop is assumed to hit first (pessimistic).
- **Walk-forward:** 8 years of training $\rightarrow$ 3 years of testing, sliding anchor. The test parameters come only from the training window. The concatenation of the test segments is the honest result.

3. How to read the Volume Profile

The profile is a histogram of volume by price. Where there was a lot of volume (HVN), price 'sticks' — it is accepted value; where there was little (LVN), price 'slips'. The POC is the price with the most volume; the Value Area (~70% of volume, ± 1 standard deviation of a Gaussian) runs from the VAL to the VAH. These levels are objective — better than support/resistance drawn 'by eye'.

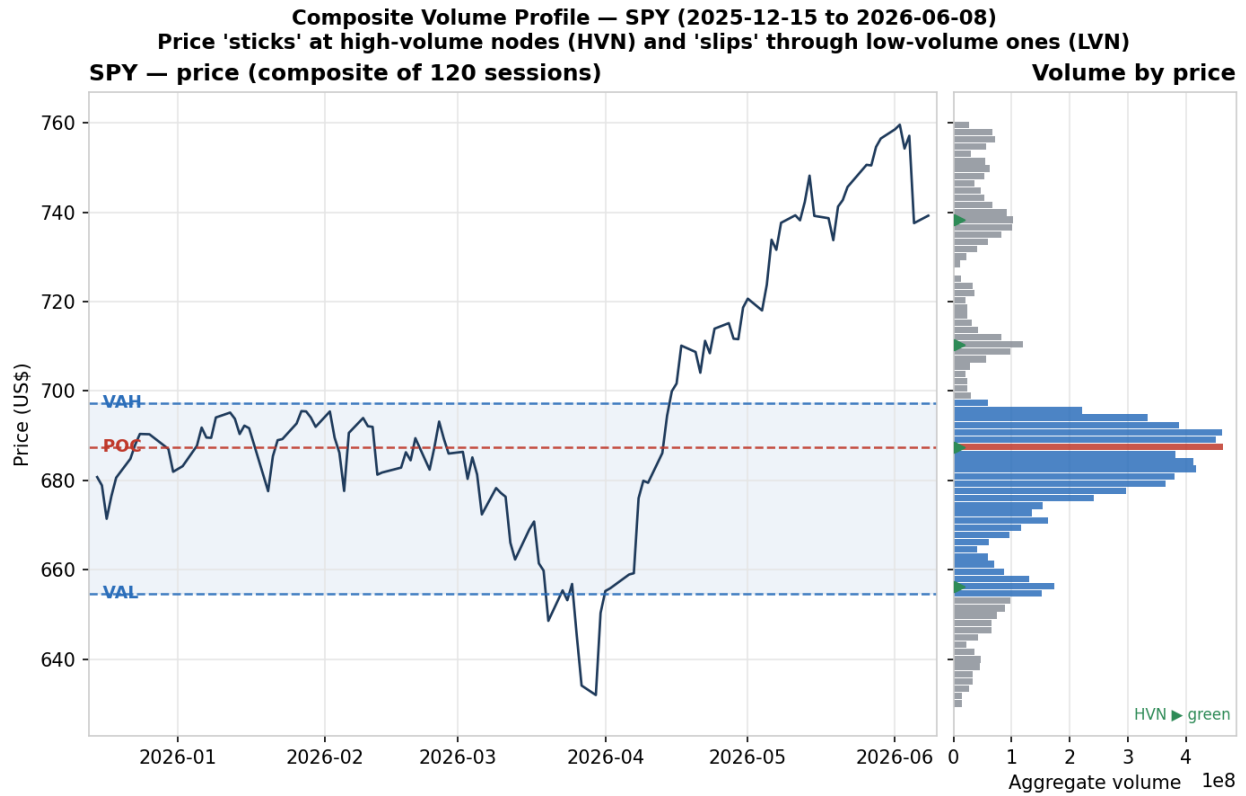


Figure 1 — Composite profile of SPY (120 sessions): the volume cluster at the POC coincides with the band where price spent the most time.

4. Results: what works (and where)

4.1 Out-of-sample overview

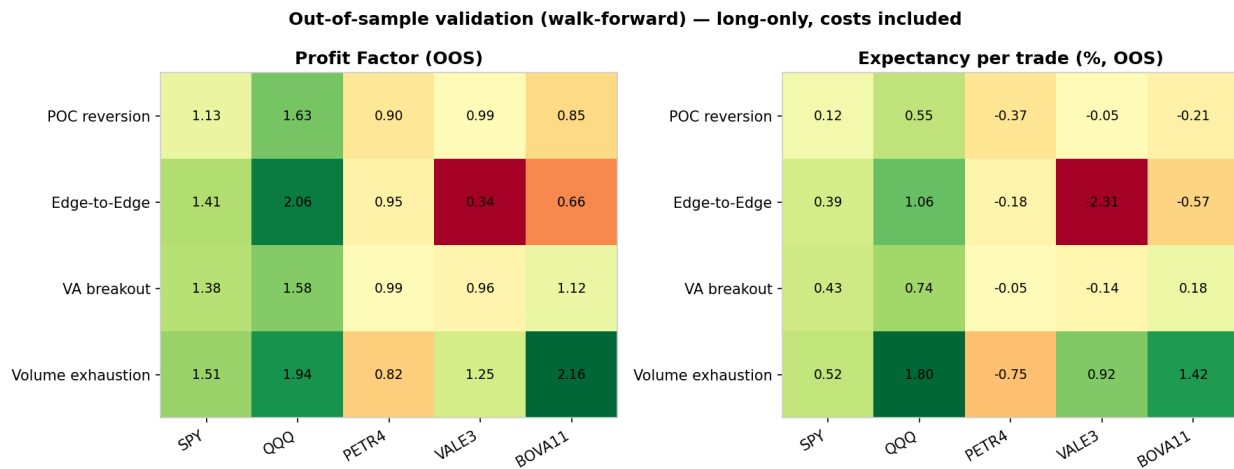


Figure 2 — Walk-forward validation (long-only, with costs). Green = positive edge. The pattern is clear: US green, Brazilian stocks red.

4.2 Full out-of-sample performance table

Strategy	Asset	N	Win	Exp/trade	PF	CAGR	Sharpe	MaxDD
POC reversion	SPY	97	44%	0.12%	1.13	-0.0%	0.00	-12.5%
POC reversion	QQQ	98	47%	0.55%	1.63	1.2%	0.39	-5.9%
POC reversion	PETR4	35	46%	-0.37%	0.90	-0.2%	-0.16	-6.9%
POC reversion	VALE3	27	33%	-0.05%	0.99	-0.2%	-0.12	-11.9%
POC reversion	BOVA11	53	42%	-0.21%	0.85	-0.7%	-0.21	-9.0%
Edge-to-Edge	SPY	109	36%	0.39%	1.41	0.9%	0.30	-9.2%
Edge-to-Edge	QQQ	79	43%	1.06%	2.06	1.8%	0.57	-6.8%
Edge-to-Edge	PETR4	42	31%	-0.18%	0.95	-0.3%	-0.15	-9.3%
Edge-to-Edge	VALE3	25	12%	-2.31%	0.34	-1.0%	-0.66	-16.5%
Edge-to-Edge	BOVA11	48	27%	-0.57%	0.66	-1.7%	-0.39	-19.7%
VA breakout	SPY	188	41%	0.43%	1.38	1.0%	0.26	-12.2%
VA breakout	QQQ	179	44%	0.74%	1.58	2.8%	0.60	-10.4%
VA breakout	PETR4	96	32%	-0.05%	0.99	-1.0%	-0.23	-21.4%
VA breakout	VALE3	108	36%	-0.14%	0.96	-0.4%	-0.09	-15.9%
VA breakout	BOVA11	86	44%	0.18%	1.12	0.4%	0.11	-12.5%
Volume exhaustion	SPY	74	47%	0.52%	1.51	0.1%	0.09	-7.7%
Volume exhaustion	QQQ	41	54%	1.80%	1.94	0.6%	0.39	-4.6%
Volume exhaustion	PETR4	103	38%	-0.75%	0.82	-0.8%	-0.27	-16.9%
Volume exhaustion	VALE3	89	40%	0.92%	1.25	-0.2%	-0.05	-13.8%
Volume exhaustion	BOVA11	35	49%	1.42%	2.16	2.1%	0.66	-8.4%

Table 1 — Out-of-sample performance by strategy × instrument. Profit Factor and expectancy color-coded (green = profitable).

4.3 Out-of-sample equity curves

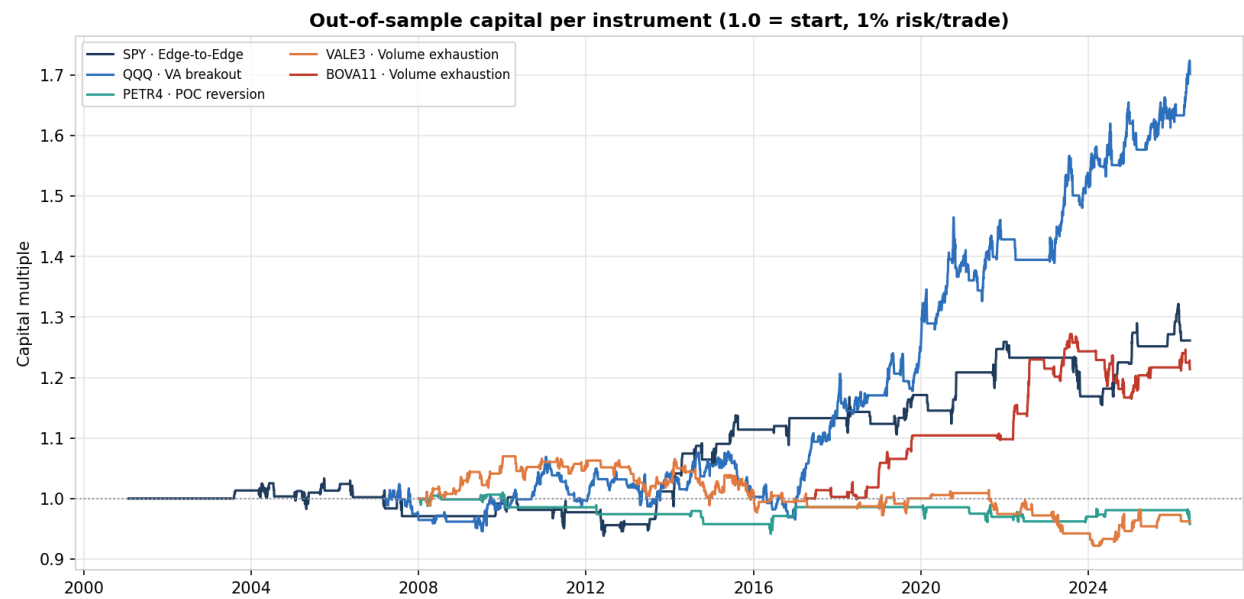


Figure 3 — OOS equity of the best long strategy per instrument. QQQ (breakout) leads; Brazilian stocks stay below 1.0.

4.4 The edge is not overfitting

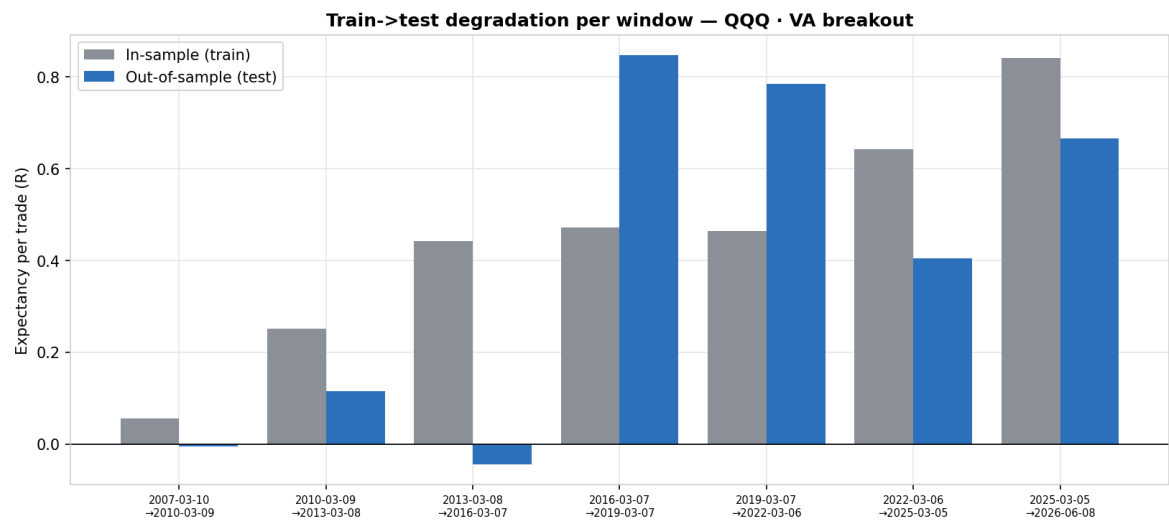


Figure 4 — Training vs test expectancy per window. Out-of-sample performance tracks (and sometimes exceeds) the in-sample — a sign of a genuine edge, not overfitting.

5. Flagship: Volume exhaustion

The tactic of buying selling exhaustion (new lows on no volume) was the most robust. Below, the parameter sensitivity and the return distribution on SPY.

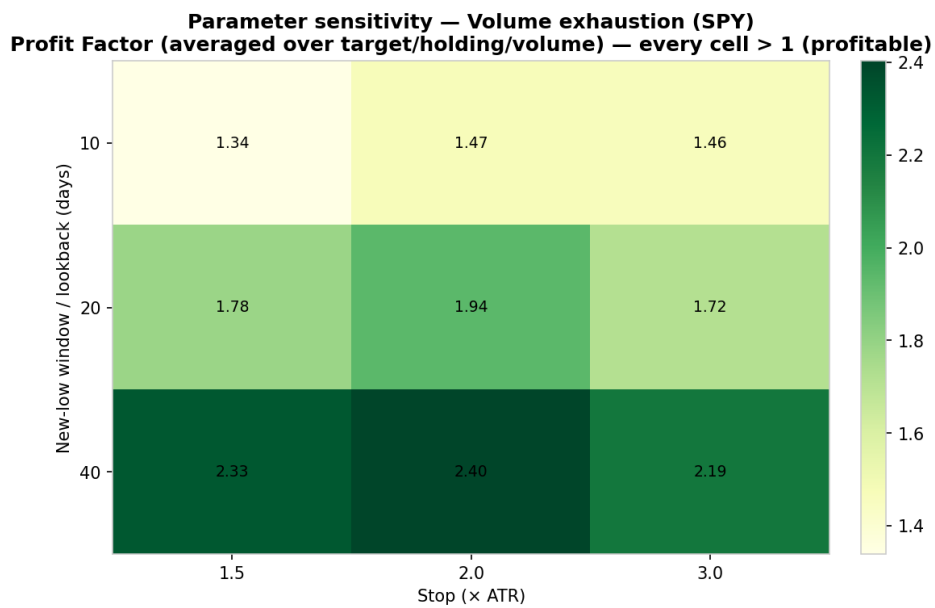


Figure 5 — Profit Factor by window $\times$ stop. Larger windows (40 days) and a stop of $\sim 2 \times \text{ATR}$ were the strongest; all cells are profitable.

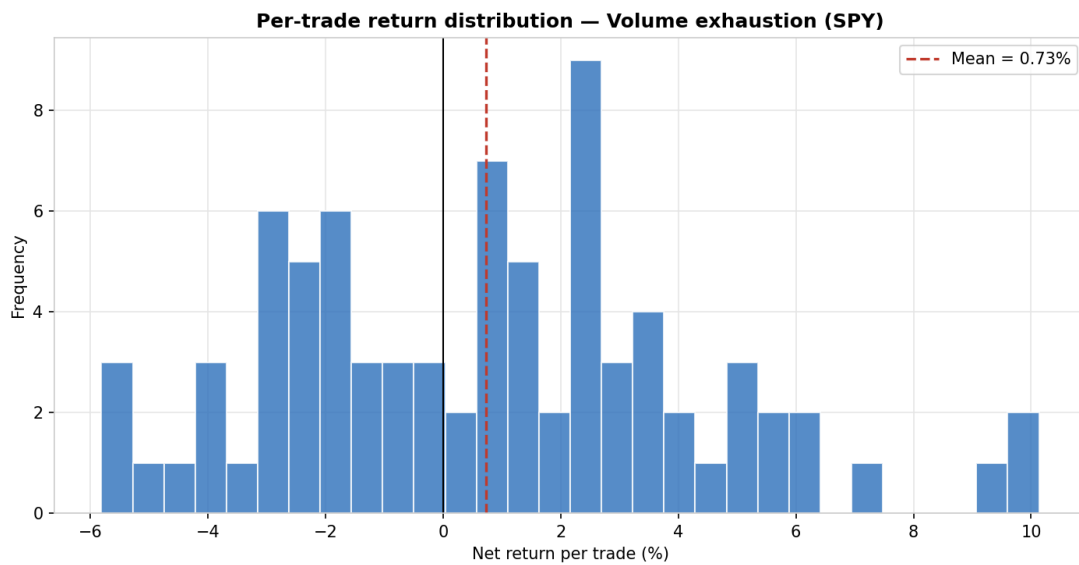


Figure 6 — Per-trade return distribution: favorable skew (losses capped by the stop, a few large gains).

6. The method's claims vs the data

6.1 Volume reading: divergence / 'no-supply'

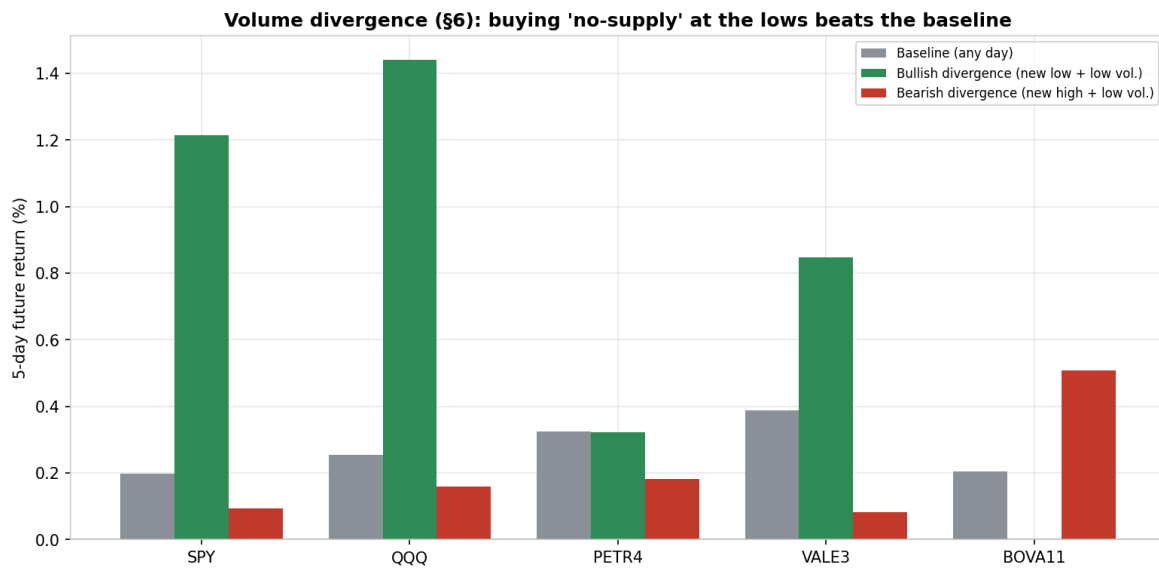


Figure 7 — 5-day forward return. Buying new lows on low volume (bullish divergence) beats the baseline on SPY, QQQ and VALE3 — the volume reading of \$6 has merit.

6.2 Day-types do not predict continuation

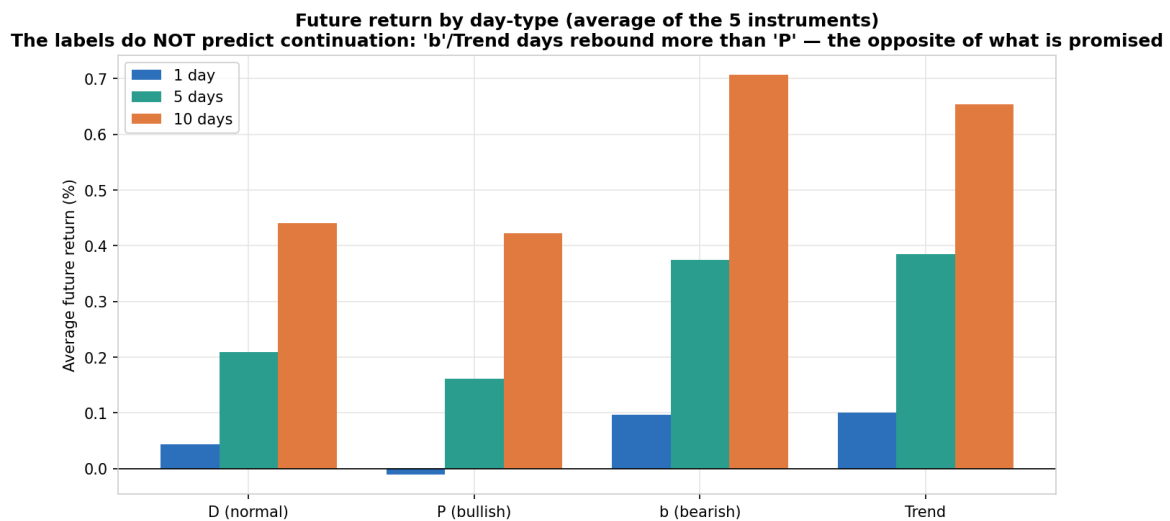
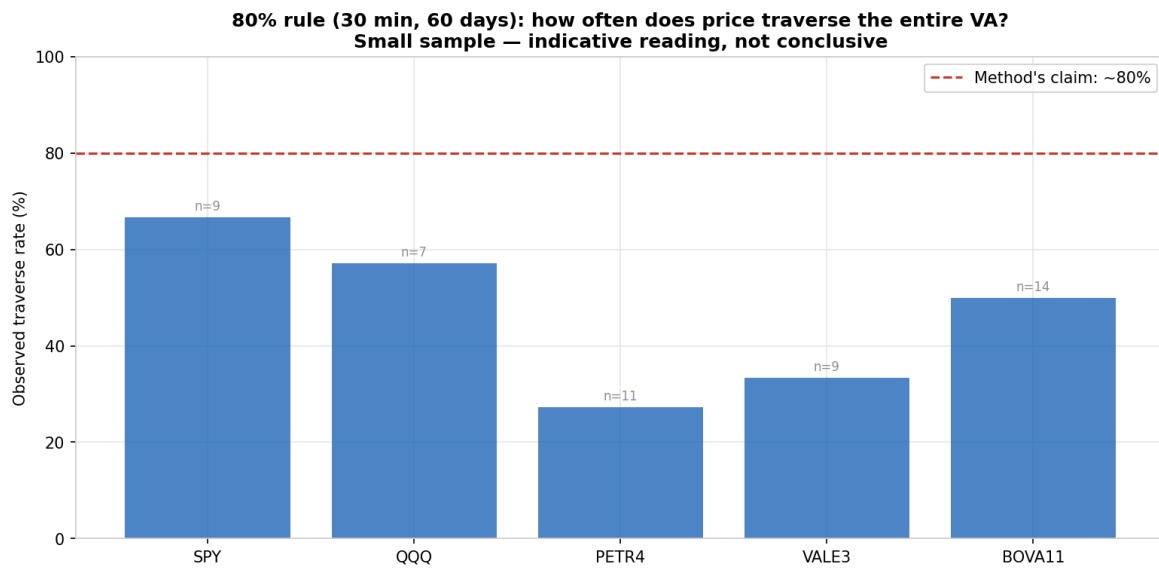


Figure 8 — Forward return by day-type (average across the 5 assets). If 'P' were bullish and 'b' bearish, the bars would diverge in that direction. The opposite happens (reversion), refuting the use of the labels as a continuation signal.

6.3 The 80% Rule does not traverse 80%



Asset	Setups	Traverse rate	Win rate	Exp/trade	Verdict
SPY	9	67%	44%	-0.08%	negative
QQQ	7	57%	57%	-0.18%	negative
PETR4	11	27%	45%	-0.20%	negative
VALE3	9	33%	11%	-0.34%	negative
BOVA11	14	50%	21%	-0.25%	negative

Table 2 — 80% Rule (30 min, 60 days). Traverse rate well below 80% and negative expectancy after costs. **Strong caveat:** tiny sample (7-14 setups/asset) — indicative, not conclusive.

7. Diversified portfolio

Each strategy in isolation trades little (low exposure). Combining, for each instrument, the strategy that validated OOS into an equal-capital portfolio raises exposure and diversifies risk — the result is a remarkably smooth return stream.

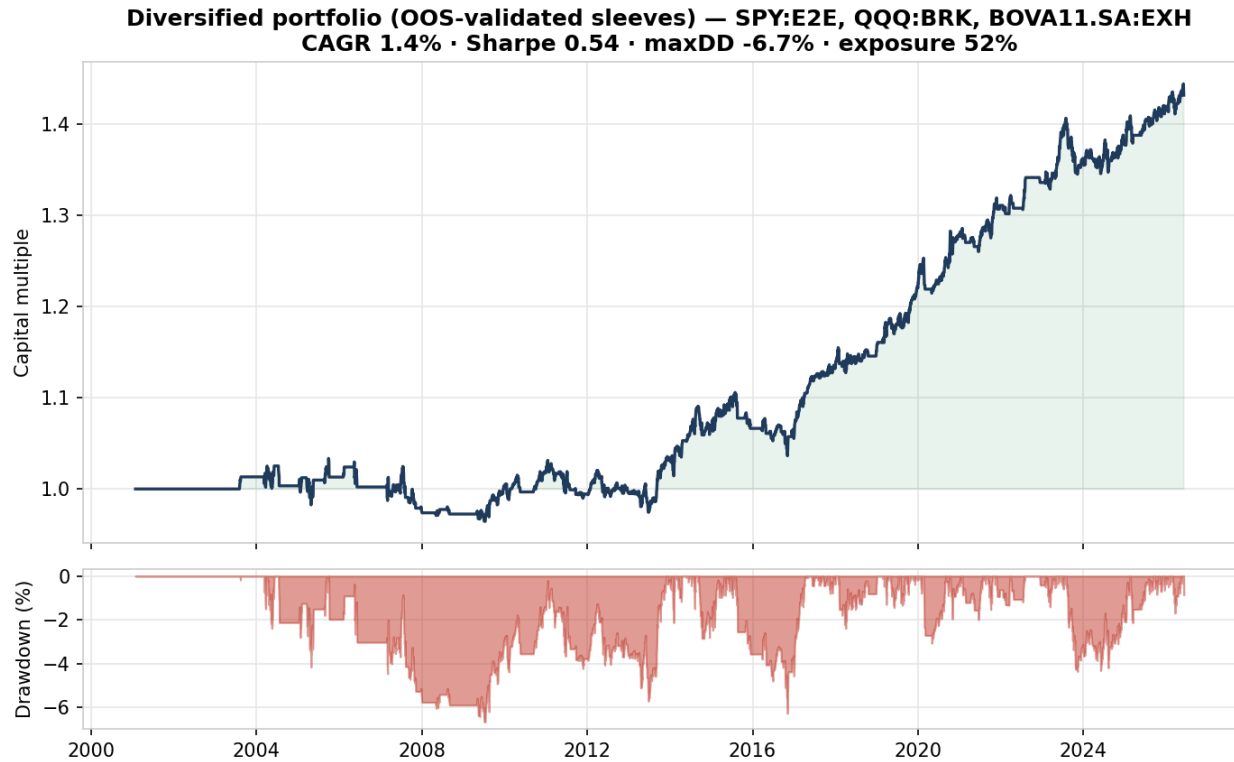


Figure 9 — Portfolio (OOS-validated sleeves) and its drawdown. Stable curve, shallow drawdown.

Portfolio	Sleeves	CAGR	Sharpe	MaxDD	Exposure
OOS-positive sleeves only	3	1.4%	0.54	-6.7%	52%
All 5 instruments	5	0.8%	0.37	-6.5%	60%

Table 3 — Portfolio metrics (1% risk/trade, no leverage).

8. Conservative vs aggressive: the risk dial

Two axes control the risk profile: (a) the **parameters** (wide stop and high selectivity = more conservative) and (b) the **sizing/leverage**. Since Sharpe is practically invariant to sizing, leveraging only trades return for drawdown — it does not create edge. The choice is one of risk appetite, not of quality.

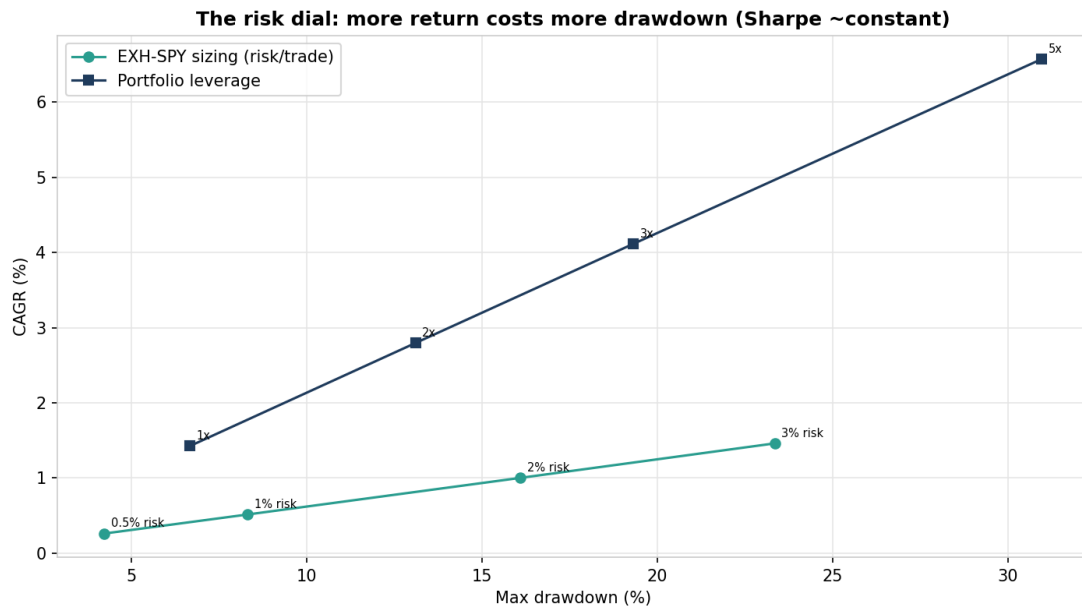


Figure 10 — CAGR vs drawdown as risk/leverage varies. The portfolio line dominates the single-asset one (diversification).

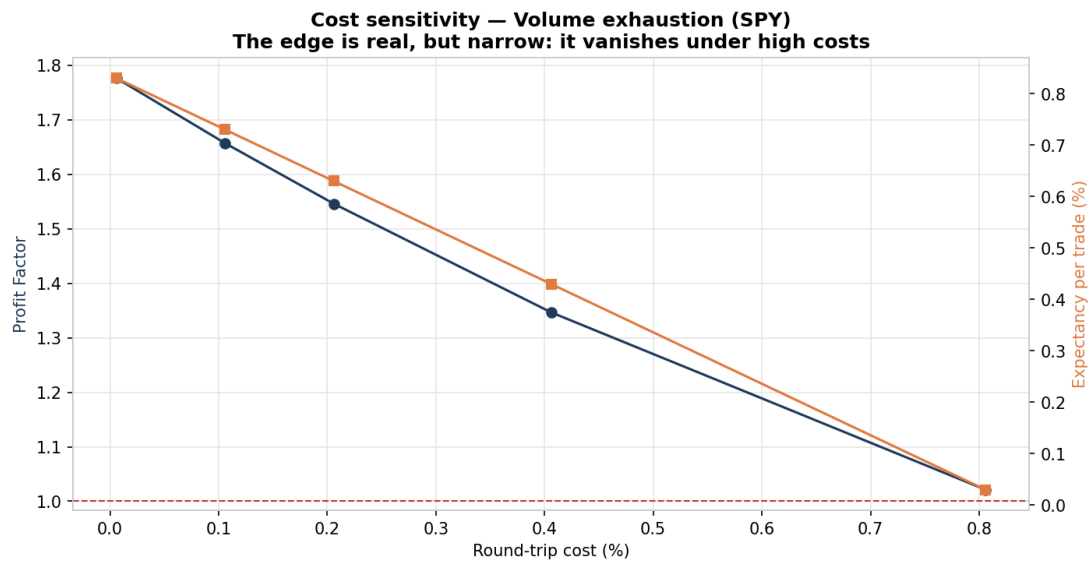
Parameter profile	N	Win	Exp/trade	PF	MaxDD
Conservative (wide stop, VA 0.80)	78	58%	0.43%	1.31	-7.0%
Aggressive (tight stop, 5xATR target)	81	37%	0.24%	1.16	-9.0%

Table 4 — Parameter profiles (Exhaustion, SPY). Conservative = higher win rate; aggressive = higher payoff, lower win rate.
The classic tradeoff.

Leverage	CAGR	Sharpe	MaxDD	Total return
1x	1.4%	0.54	-6.7%	43.1%
2x	2.8%	0.54	-13.1%	101.2%
3x	4.1%	0.54	-19.3%	177.9%
5x	6.6%	0.54	-31.0%	402.3%

Table 5 — Portfolio under leverage. Sharpe constant; only the scale of return and drawdown changes.

9. Cost sensitivity and context



Round-trip cost	Expectancy/trade	Profit Factor	CAGR
0.01%	0.830%	1.78	0.54%
0.11%	0.730%	1.66	0.45%
0.21%	0.630%	1.55	0.36%
0.41%	0.430%	1.35	0.17%
0.81%	0.030%	1.02	-0.19%

Table 6 / Figure 11 — The edge is real but narrow: it approaches breakeven (PF 1.0) under ~0.8% round-trip costs. Cheap execution is a prerequisite.

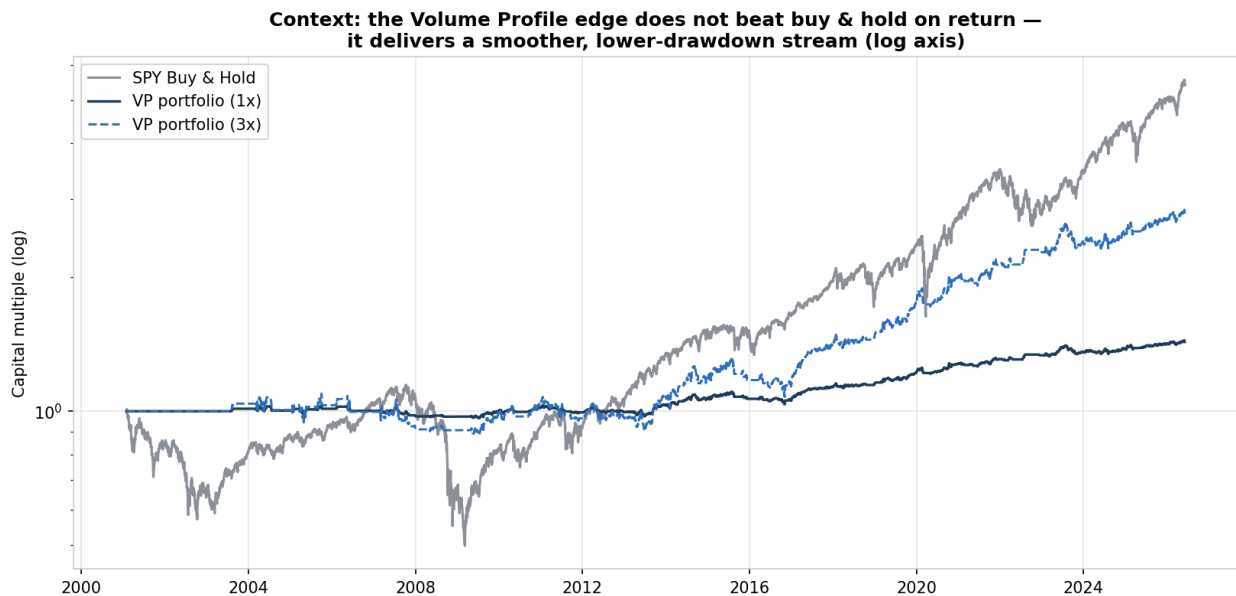


Figure 12 — Context (log axis): SPY buy & hold wins on absolute return, but with much larger drawdowns (note 2008). The VP portfolio delivers less return and far less suffering.

10. Complementary studies (§4.3, §6, §7)

To exhaust the reference document, we also tested the 'raw' volume readings (absorption and healthy movement), the effect of the 'signal candle' and the sensitivity to histogram resolution.

10.1 Raw volume readings: absorption and healthy movement

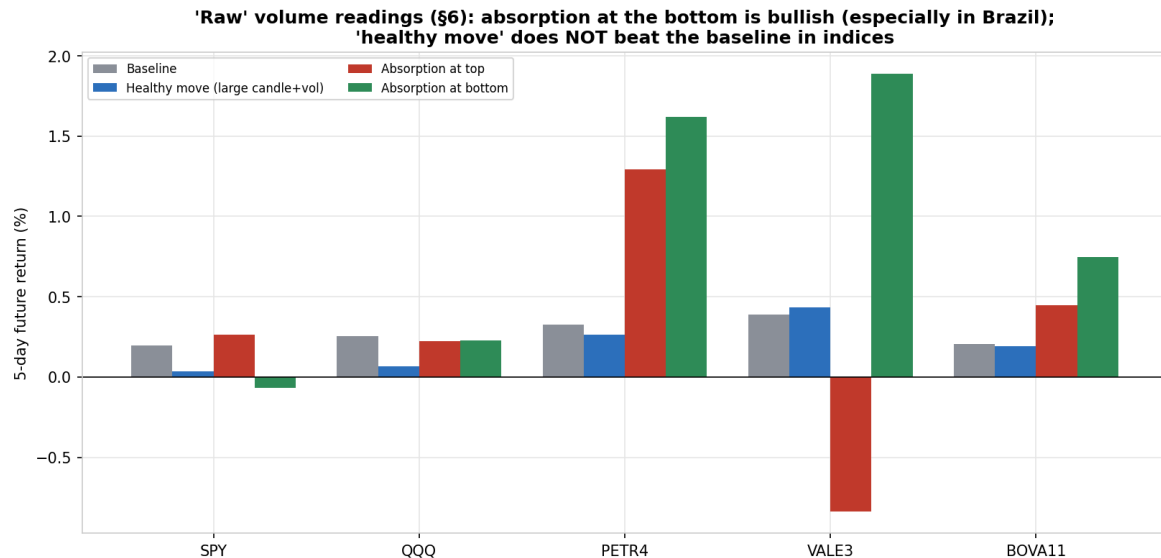


Figure 13 — **Absorption at the bottom** (new low + huge volume + small body) precedes strong bounces, especially in Brazilian stocks (VALE3 +1.9%, PETR4 +1.6% vs ~0.35% baseline). The 'healthy movement' (large candle and volume), by contrast, does **not** beat the baseline on indices — effect $\neq$ outcome, contrary to what was promised.

10.2 The 'signal candle' (volume confirmation) has merit

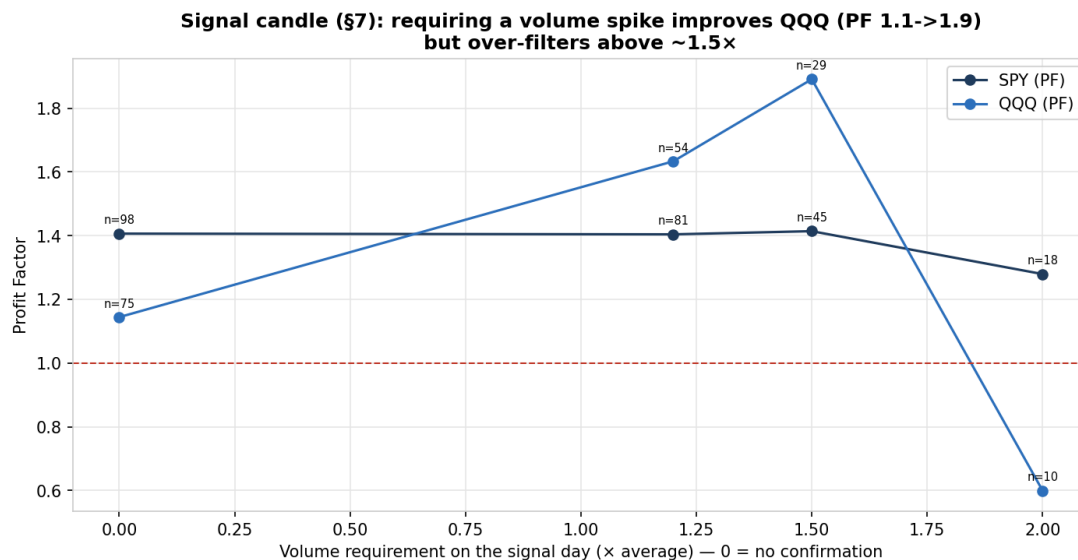


Figure 14 — Requiring a volume spike on the signal day (§7) clearly improves QQQ (Profit Factor 1.1 $\rightarrow$ 1.9 at ~1.2-1.5x the average), but over-filters above that. It confirms the recommendation to 'wait for the volume candle' — in moderation.

10.3 Profile resolution (the '400 rows' rule)

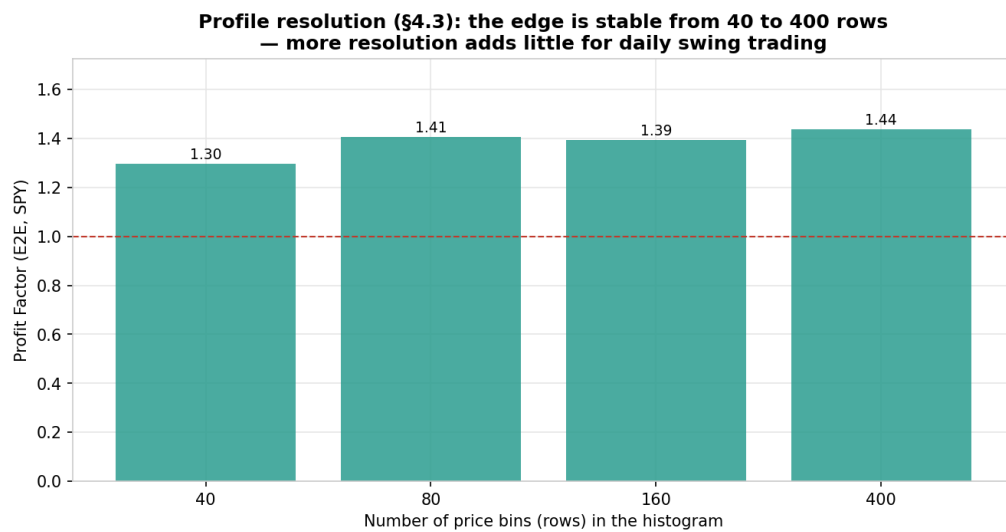


Figure 15 — The edge is stable from 40 to 400 price bins; more resolution helps only marginally in daily swing trading. The insistence on '400 rows' matters more in fine-grained intraday than here.

11. Falsification and ablation tests: is the edge real?

Before trusting the Volume exhaustion sleeve, we subjected it to five falsification tests with a **fixed** config (no re-optimization, fixed seed) to answer: does the edge come from **volume and structure**, or is it just **long bias / buy-the-dip** on an asset that goes up? The tests: (1) volume permutation; (2) price-only ablation; (3) random-entry control; (4) excess return over exposure and risk-free (CDI 14.5% BRL / T-bill ~2% USD); (5) 95% CI via bootstrap.

Asset	Volume shuffle	Beats price-only	Beats random long	Alpha over exposure	Beats risk-free	PF CI excludes 1	PASSES ALL
SPY	YES	YES	YES	no	no	no	no
QQQ	no	YES	no	no	no	no	no
PETR4	no	no	no	no	no	no	no
VALE3	no	no	no	no	no	no	no
BOVA11	no	YES	no	no	no	no	no

Table 8 — Verdict per sleeve. **None of the five instruments passes all six tests.** SPY is the only one showing a statistically real volume signal (passes shuffle, ablation and random entry), but it fails the economic tests.

Asset	Expectancy w/ volume	Expectancy price-only	Δ (volume adds)
SPY	0.730%	0.344%	0.386%
QQQ	0.496%	0.046%	0.450%
PETR4	-0.722%	-0.527%	-0.195%
VALE3	-0.091%	0.303%	-0.394%
BOVA11	0.090%	-0.198%	0.288%

Table 9 — Price-only ablation: the volume filter nearly doubles expectancy on SPY/QQQ (more selective), but hurts on Brazilian stocks. Volume adds selectivity — on some assets.

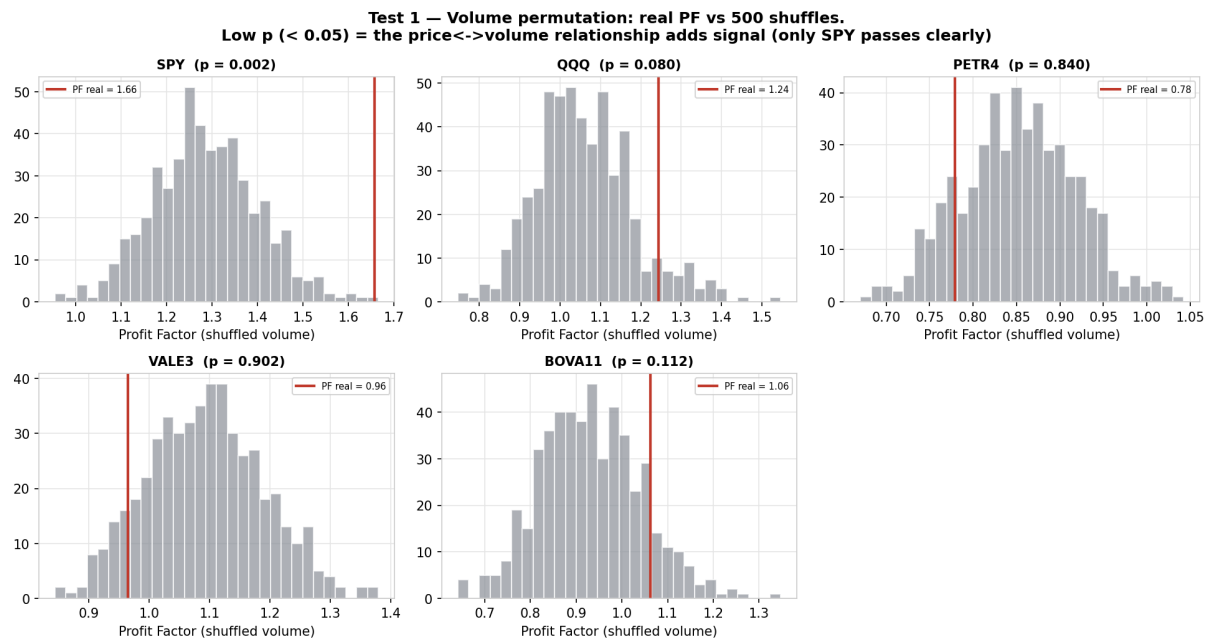


Figure 16 — Volume permutation test. Only on SPY is the real PF (red line) clearly in the right tail of the 500 shuffles ($p = 0.002$): the price $\leftrightarrow$ volume relationship carries signal. For the rest, shuffled volume does as well as the real thing.

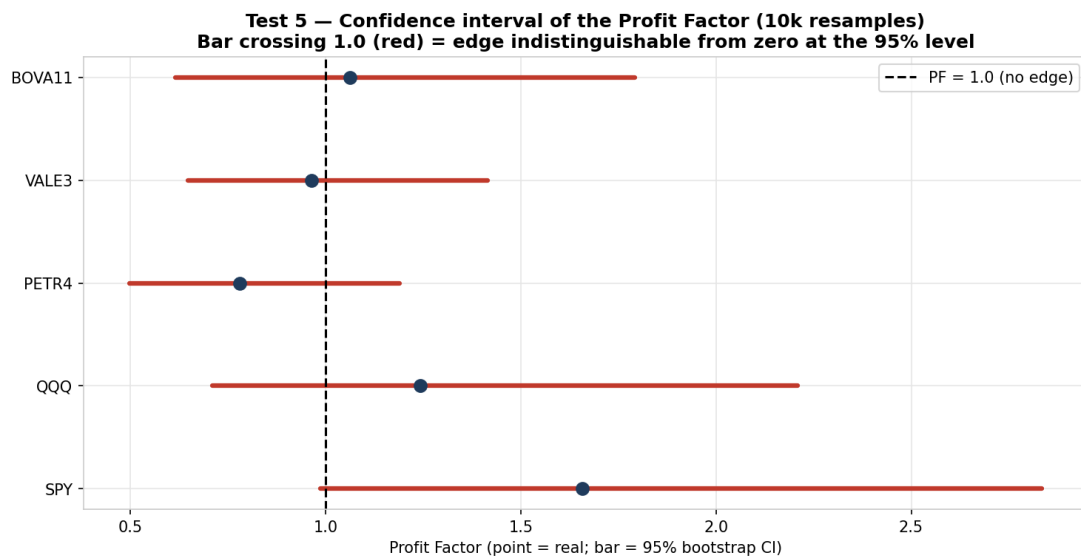


Figure 17 — 95% CI of the Profit Factor (bootstrap, 10k). All bars cross 1.0 — including SPY (lower bound 0.99). With ~80-150 trades, the edge is **not statistically distinguishable from zero** at 95%.

Test 3 — Long-bias control: real return vs 500 random entries of the same size/holding.
The strategy must sit to the right of the mass ($p < 0.05$) to claim real timing

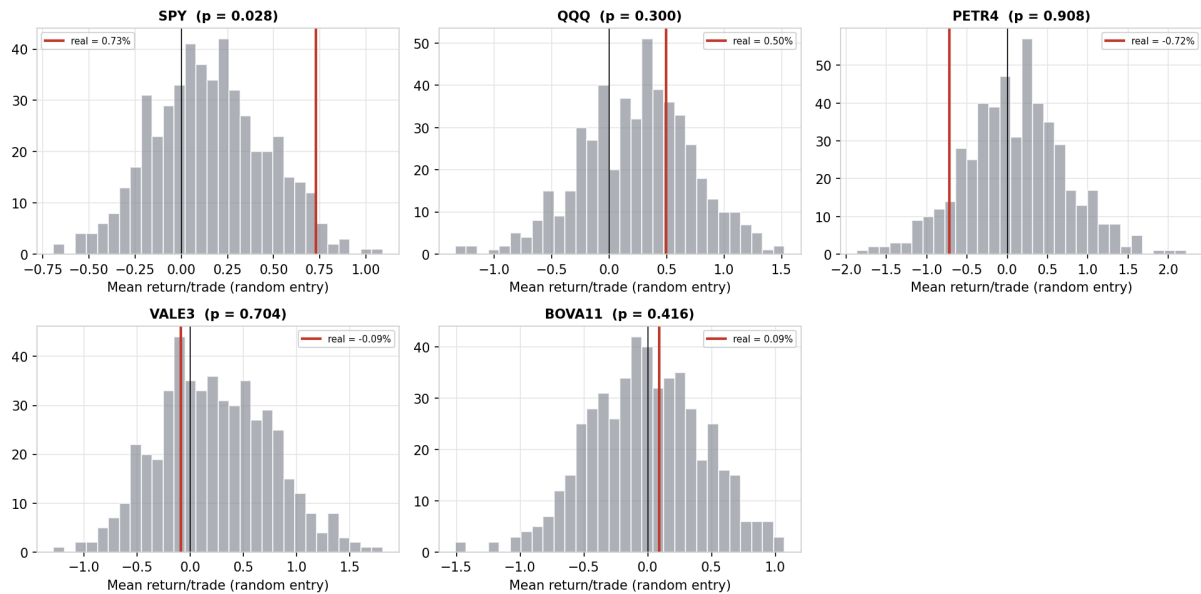


Figure 18 — Random-entry control. Only SPY beats the 'random long' ($p = 0.028$); QQQ/BR do not — their return is consistent with simply being long on arbitrary dates.

Honest verdict. The test is relentless and instructive: the apparent gain of the strategies is, for the most part, **long exposure** to assets that went up — not a robust volume edge. The volume filter *adds selectivity* (positive ablation on SPY/QQQ) and, on SPY, the price↔volume relationship is *statistically real*. But no sleeve generates alpha over its own exposure, none beats the risk-free rate at 1% risk, and none has a PF whose 95% CI excludes 1.0. Conclusion: at 1% risk, as a standalone strategy, **there is no demonstrable economic edge**; what remains (on SPY) is a small, real, but statistically fragile signal at the available sample size.

12. Parameter recommendations

Based on the sweep and the out-of-sample validation, these are the variables that behaved best. **Important (in light of §11):** these recommendations describe the *least bad* and most historically robust configuration — not a proven economic edge. Treat them as a starting point for study and paper trading, not as a system ready for real capital.

Variable	Conservative profile	Aggressive profile
Instruments	Liquid index ETFs only (SPY, QQQ)	ETFs + volatile names (VALE3) with breakout
Strategy	Volume exhaustion / Edge-to-Edge (long)	VA breakout + Exhaustion (long)
Direction	Long only	Long (shorts lost on everything)
Profile window/lookback	40 days	20 days
Value Area	0.80 (more selective)	0.70
Stop	$\approx 2.5\text{-}3.0 \times \text{ATR}$ (wide)	$\approx 1.5 \times \text{ATR}$ (tight)
Target	POC / opposite edge	$5 \times \text{ATR}$ (let it run)
Maximum holding	10 days	20 days
Trend filter	On (does not fade trend)	On for breakout
Risk per trade	0.5-1.0% of capital	2-3% of capital
Leverage	1× (none)	up to 3× in the diversified portfolio

Table 7 — Recommended variables. **Rule of thumb:** the conservative profile prioritizes low drawdown and robustness across assets; the aggressive one accepts larger drawdown for more CAGR, with no illusion of superior Sharpe.

What improved results the most (in order of impact): (1) trading long only on liquid indices; (2) using the volume reading (exhaustion/absorption at the bottom) instead of the raw profile; (3) the trend filter, which cut the drawdown from -74% to -12% on reversion; (4) longer windows (40 days) on the exhaustion signal; (5) confirming with a volume spike (signal candle) — which raised QQQ's PF from 1.1 to 1.9; (6) diversifying into a portfolio.

13. Limitations and honest conclusion

- **Profile approximation:** without long tick/intraday data, volume is distributed uniformly across the daily range. This is standard practice, but it is an approximation of the 'real' profile.
- **The 80% Rule and intraday day-types** were tested on a short sample (~60 days of 30-min bars) — an indicative reading. A definitive test requires paid historical intraday data.
- **Small, cost-sensitive edge:** it survives OOS but evaporates under high costs. It requires cheap execution and discipline; it does not tolerate over-trading.
- **Survivorship and regime:** 33 years include several regimes, but the future may differ. Walk-forward mitigates, but does not eliminate, this risk.
- **Day-trading base rate (USP/FGV):** of those who persisted for +300 days, 97% lost money and there was no evidence of learning. A statistical edge in a backtest $\neq$ guaranteed profit for a real trader.

Conclusion. Volume Profile is neither magic nor charlatanism. It is a legitimate auction lens (why price sticks where there is volume) and provides objective levels, and the volume filter demonstrably adds selectivity. But the most honest part of this study is §11: under rigorous falsification, what looked like an edge is, to a large extent, **long exposure** to assets that went up. Only on SPY is the volume signal statistically real — and even there it generates no alpha over its own exposure, does not beat the risk-free rate at 1% risk, and its Profit Factor is not distinguishable from 1.0 at 95% confidence. **We found no standalone, robust economic edge.** This does not invalidate Volume Profile as a tool for context and risk management — it invalidates the promise of easy profit. The metric that matters is expectancy after costs, validated against chance; and the evidence calls for skepticism, not faith.

Generated by a reproducible pipeline (Python). Code and data are version-controlled. Not investment advice.